

Using MongoDB to Improve the IT Performance of an Enterprise

This article targets developers and architects who are looking for open source adoption in their IT ecosystems. The authors describe an actual enterprise situation, in which they adopted MongoDB in their work flow to speed up processes.



In the last decade or so, the amount of data generated has grown exponentially. The ways to store, manage and visualise data have shifted from the old legacy methods to new ways. There has been an explosion in the number and variety of open source databases. Many are designed to provide high scalability, fault tolerance and have core ACID database features. Each open source database has some special features and, hence, it is very important for a developer or any enterprise to choose with care and analyse each specific problem statement or use case independently. In this article, let us look at one of the open source databases that we evaluated and adopted in our enterprise ecosystem to suit our use cases.

MongoDB, as defined in its documentation, is an open source, cross-platform, document-oriented database that provides high performance, high availability and easy scalability.

MongoDB works with the concept of collections, which you can associate with the table in an RDBMS like MySQL and Oracle. Each collection is made up of documents (like XML, HTML or JSON), which are the core

entity in MongoDB and can be compared to a logical row in Oracle databases.

MongoDB has a flexible schema as compared to the normal Oracle DB. In the latter we need to have a definite table with well-defined columns and all the data needs to fit the table row type. However, MongoDB lets you store data in the form of documents, in JSON format and in a non-relational way. Each document can have its own format and structure, and be independent of others. The trade-off is the inability to perform joins on the data. One of the major shifts that we as developers or architects had to go through while adopting Mongo DB was the mindset shift — of getting used to storing normalised data, getting rid of redundancy in a world where we need to store all the possible data in the form of documents, and handling the problems of concurrency.

The horizontal scalability factor is fulfilled by the ‘sharding’ concept, where the data is split across different machines and partitions called shards, which help further scaling. The fault tolerance capabilities are enabled by replicating data on different machines or data centres, thus making the data available in case of server failures. Also,